

# Lahore Garrison University

DHA Phase 6 Lahore

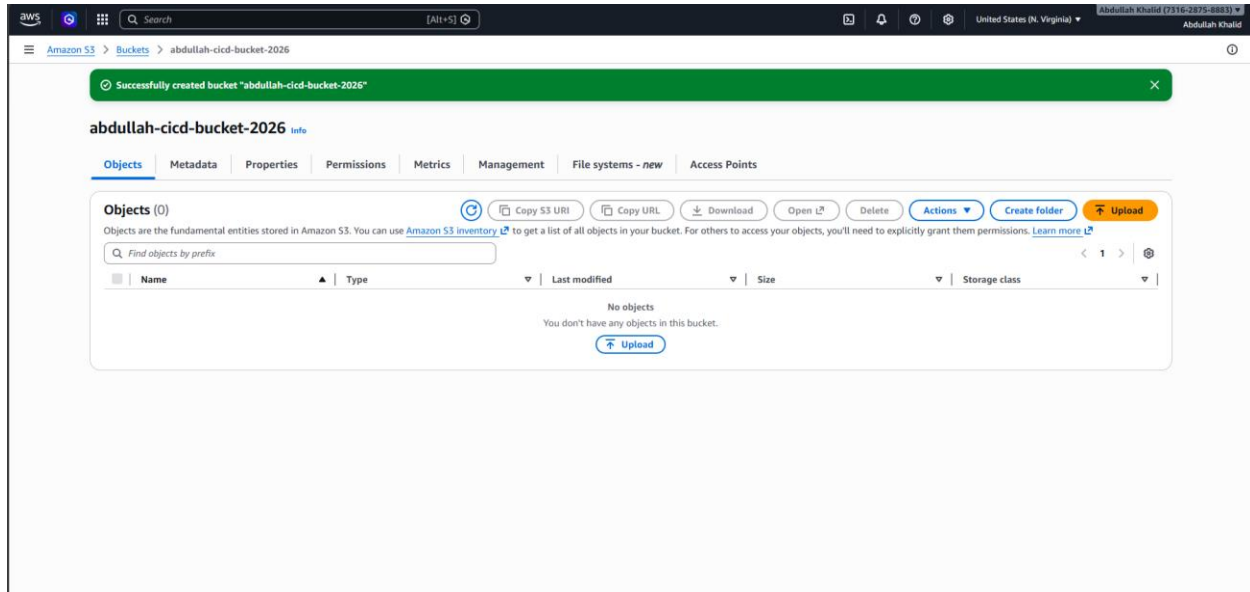


|                        |                        |
|------------------------|------------------------|
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| <b>Section</b>         | <b>E</b>               |
| <b>Subject</b>         | <b>Cloud Computing</b> |
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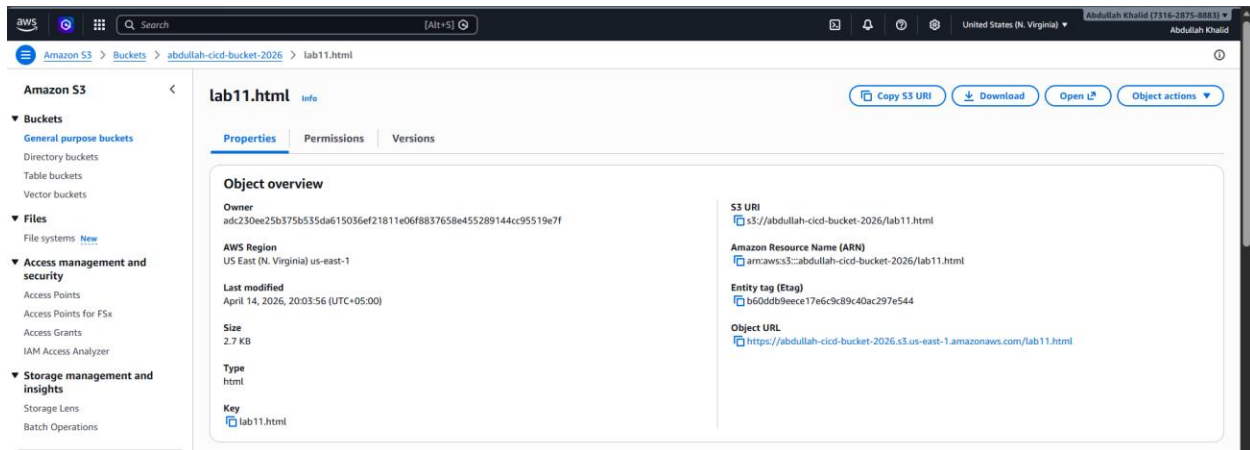
## LAB 7

### CI/CD using AWS CodePipeline

S3 bucket created:



S3 bucket Details for commit code:



## Q1 # Pushing Code to observe pipeline behavior:

The screenshot shows the AWS CodeCommit console interface. A green notification bar at the top states "lab11.html has been committed to main" with a "View commit: 6bd1b86d" link. The left sidebar contains navigation options under "Developer Tools" and "CodeCommit", including "Source", "Getting started", "Repositories", "Code", "Pull requests", "Commits", "Branches", "Git tags", "Settings", "Approval rule templates", "Artifacts", "Build", "Deploy", "Pipeline", and "Settings". The main content area displays the "cicd-lab-repo" repository. The "Reference" dropdown is set to "main". Below the repository name, the file "cicd-lab-repo / lab11.html" is shown with an "Info" link and an "Edit" button. The code content is an HTML file with the following structure:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <title>Lab 11 - Canvas & SVG</title>
6
7 <style>
8   body {
9     font-family: Arial, sans-serif;
10    display: flex;
11    justify-content: center;
12    align-items: center;
13    flex-direction: column;
14    background: #f4f4f4;
15    height: 100vh;
16    margin: 0;
17  }
18
19  h2 {
20    margin-top: 20px;
21    color: #333;
22  }
23
24  #drawing-area {
25    border: 3px solid #333;
26    border-radius: 10px;
27    background: white;
28    cursor: crosshair;
29    margin-bottom: 15px;
30  }
```

The screenshot shows the AWS CodeCommit console interface. A green notification bar at the top states "buildspec.yml has been committed to main" with a "View commit: 216c19c3" link. The left sidebar is identical to the previous screenshot. The main content area displays the "cicd-lab-repo" repository. The "Reference" dropdown is set to "main". Below the repository name, the file "cicd-lab-repo / buildspec.yml" is shown with an "Info" link and an "Edit" button. The code content is a YAML file with the following structure:

```
1 version: 0.2
2
3 phases:
4   build:
5     commands:
6       - mkdir -p output
7       - cp lab11.html output/
8
9 artifacts:
10   files:
11     - '**/*.*'
12   base-directory: output
```

## Adding build part for code:

The screenshot displays the AWS CodeBuild console interface. At the top, a green banner indicates 'Project created' for 'cicd-lab-build'. Below this, the project configuration is shown with fields for Source provider (AWS CodeCommit), Primary repository (cicd-lab-repo), Artifacts upload location, and Service role. The 'Build history' tab is selected, showing a table with columns: Build run, Status, Build number, Source version, Submitter, Duration, and Completed. The table is currently empty, displaying 'No results' and 'There are no results to display.' The left sidebar contains navigation links for Developer Tools, CodeBuild, and various integration options like Jenkins, GitHub Actions, and GitLab runners.

**Project created**  
You have successfully created the following project: cicd-lab-build

**cicd-lab-build**

**Configuration**

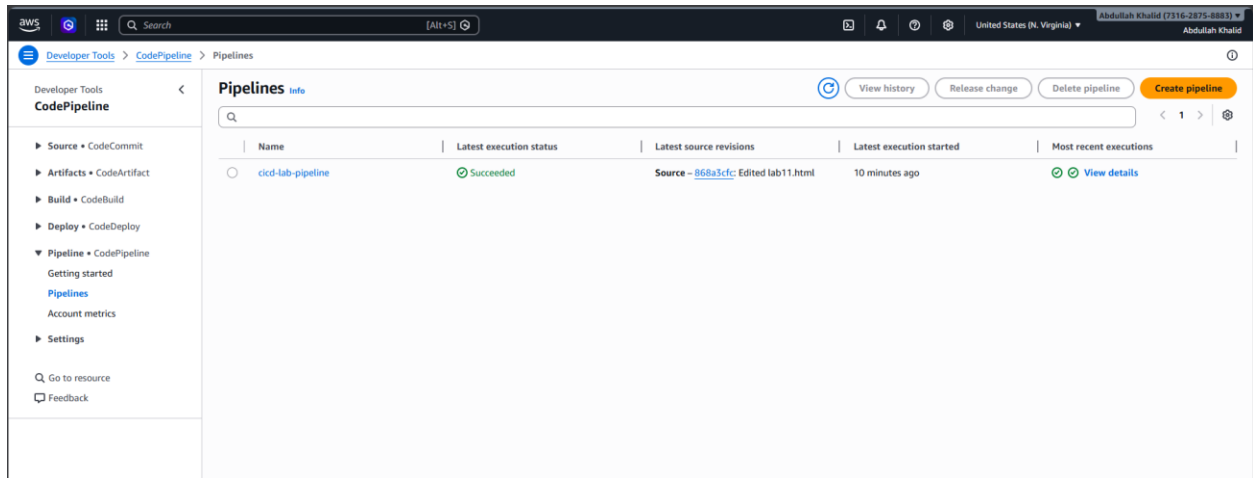
|                                   |                                     |                                |                                                                                                   |
|-----------------------------------|-------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------|
| Source provider<br>AWS CodeCommit | Primary repository<br>cicd-lab-repo | Artifacts upload location<br>- | Service role<br>arn:aws:iam::731628758883:role/service-role/codebuild-cicd-lab-build-service-role |
|-----------------------------------|-------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------|

Public builds  
Disabled

**Build history**

| Build run                                      | Status | Build number | Source version | Submitter | Duration | Completed |
|------------------------------------------------|--------|--------------|----------------|-----------|----------|-----------|
| No results<br>There are no results to display. |        |              |                |           |          |           |

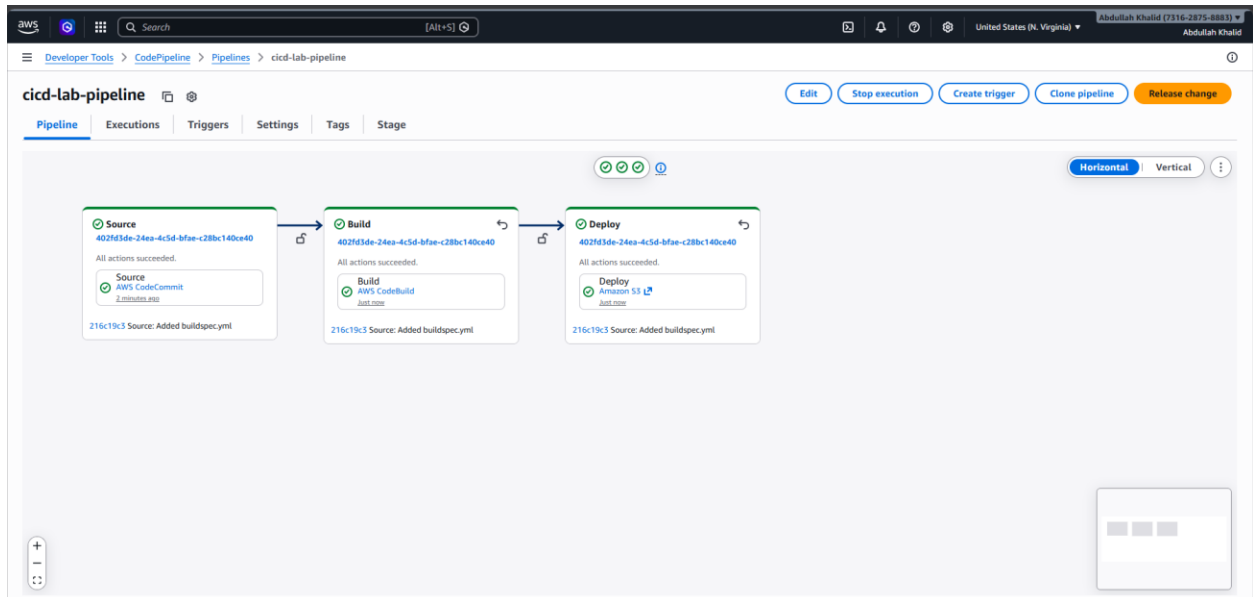
## Q2 # Pipeline Created and executed successfully:



The screenshot shows the AWS CodePipeline console. The left sidebar contains the navigation menu with options like Source, Artifacts, Build, Deploy, Pipeline, and Settings. The main area displays a table of pipelines. The table has columns for Name, Latest execution status, Latest source revisions, Latest execution started, and Most recent executions. A single pipeline named 'cicd-lab-pipeline' is listed with a 'Succeeded' status. Above the table, there are buttons for 'View history', 'Release change', 'Delete pipeline', and 'Create pipeline'.

| Name              | Latest execution status | Latest source revisions              | Latest execution started | Most recent executions       |
|-------------------|-------------------------|--------------------------------------|--------------------------|------------------------------|
| cicd-lab-pipeline | Succeeded               | Source - 868a3cfc: Edited lab11.html | 10 minutes ago           | <a href="#">View details</a> |

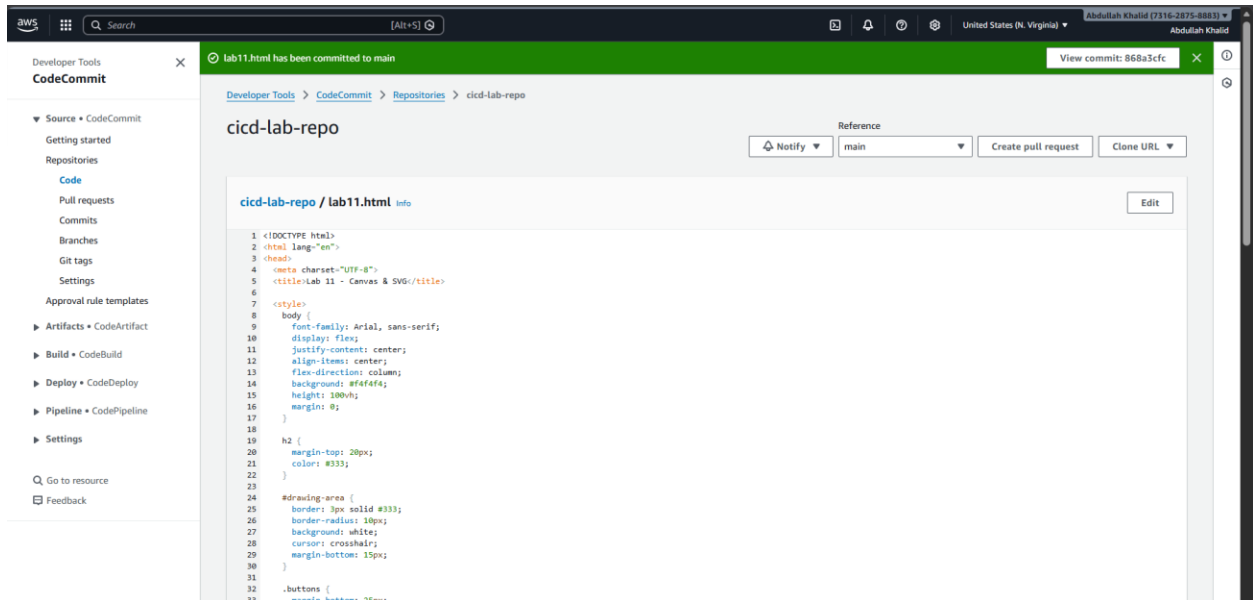
## Q2 # Source & Deploy stages in pipeline:



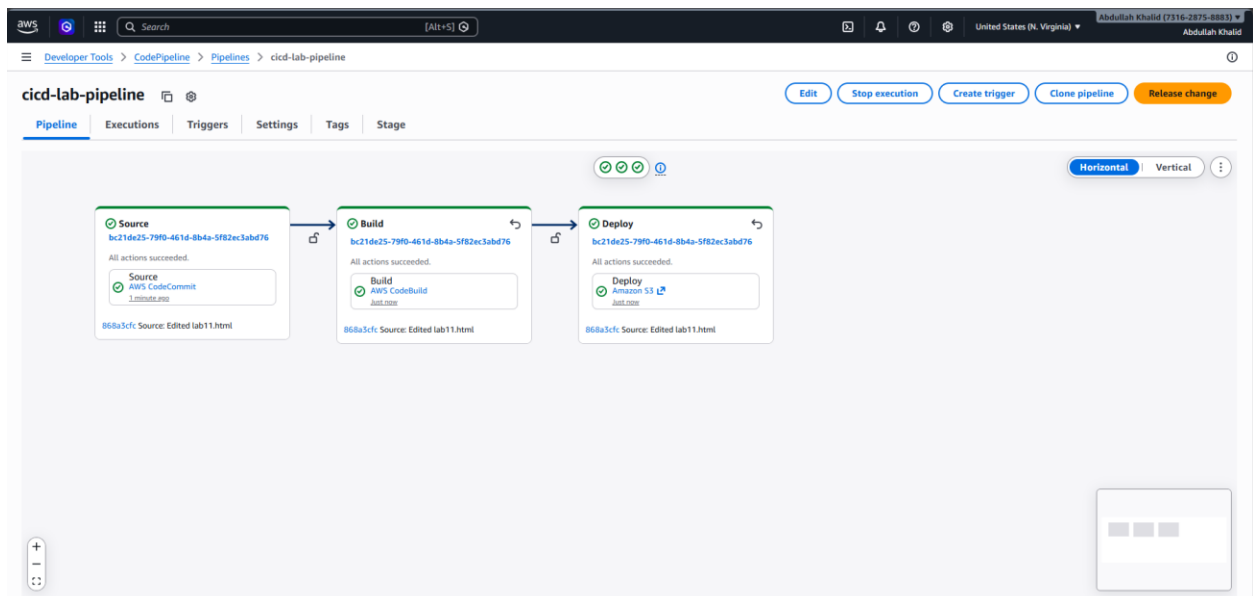
The screenshot shows the details of the 'cicd-lab-pipeline' in the AWS CodePipeline console. The top navigation bar includes buttons for 'Edit', 'Stop execution', 'Create trigger', 'Clone pipeline', and 'Release change'. Below the navigation bar, there are tabs for 'Pipeline', 'Executions', 'Triggers', 'Settings', 'Tags', and 'Stage'. The 'Pipeline' tab is selected, showing a visual representation of the pipeline stages. The pipeline consists of three stages: 'Source', 'Build', and 'Deploy'. Each stage is represented by a box containing a list of actions. The 'Source' stage has one action named 'Source' using the 'AWS CodeCommit' provider. The 'Build' stage has one action named 'Build' using the 'AWS CodeBuild' provider. The 'Deploy' stage has one action named 'Deploy' using the 'Amazon S3' provider. The pipeline is shown in a horizontal layout.

```
graph LR; Source[Source] --> Build[Build]; Build --> Deploy[Deploy];
```

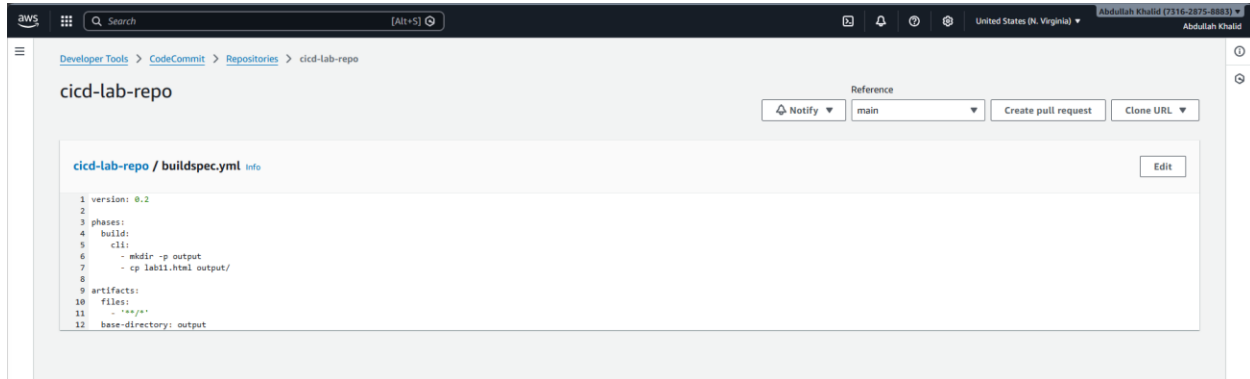
## Code is modified to analyze stages of Pipeline:



## Pipeline after modification of code:

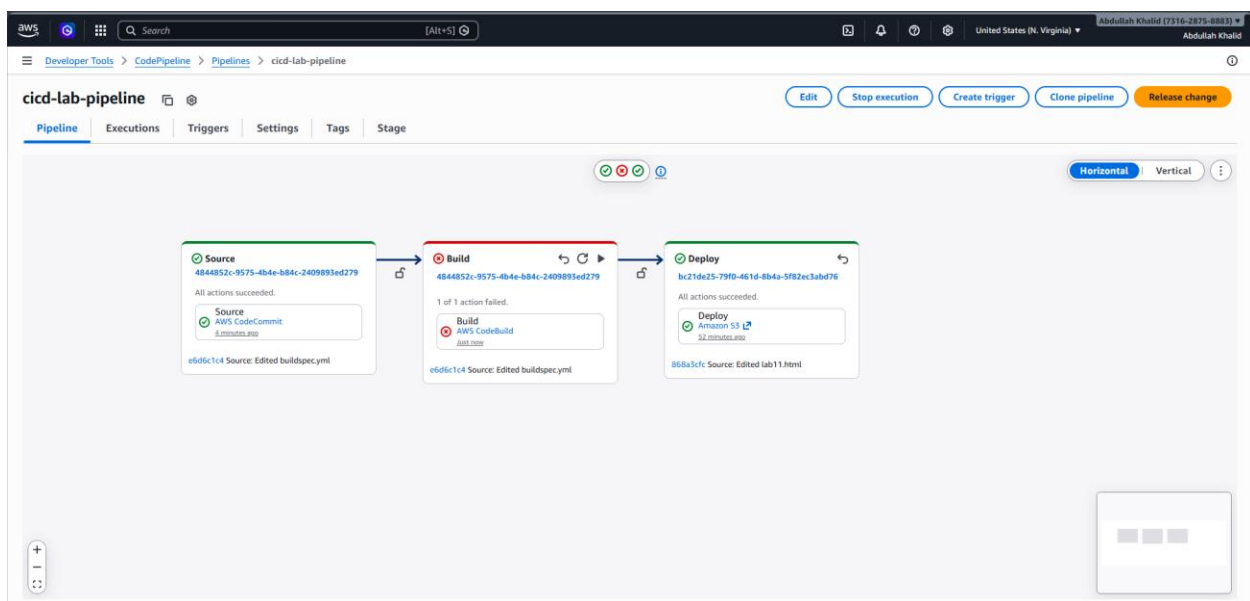


## Q3 # added bug by replacing “commands” -> “cli” for analyzing errors and logs:

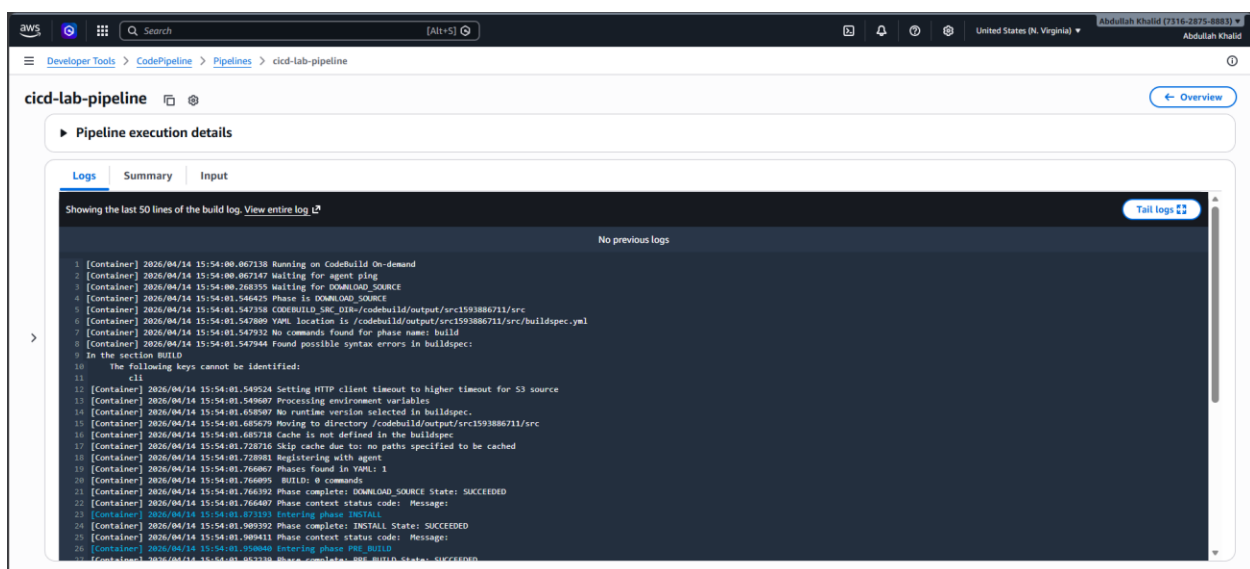


The screenshot shows the AWS CodeCommit repository page for `cicc-lab-repo`. The `buildspec.yml` file is displayed with the following content:

```
1 version: 0.2
2
3 phases:
4   build:
5     cli:
6       - mkdir -p output
7       - cp lab11.html output/
8
9 artifacts:
10  files:
11    - "**/*"
12  base-directory: output
```



The screenshot shows the AWS CodePipeline console for the `cicc-lab-pipeline`. The pipeline is in a failed state, indicated by a red circle with a white 'X'. The pipeline consists of three stages: `Source`, `Build`, and `Deploy`. The `Build` stage is highlighted in red, indicating it failed. The error message for the `Build` stage is "1 of 1 action failed." The `Deploy` stage is highlighted in green, indicating it succeeded. The pipeline is currently in a failed state, and the error message for the `Build` stage is "1 of 1 action failed."



The screenshot shows the AWS CodePipeline console for the `cicc-lab-pipeline`, specifically the `Logs` tab for the `Build` stage. The logs show the following error message:

```
1 [Container] 2026/04/14 15:54:00.067138 Running on CodeBuild On-demand
2 [Container] 2026/04/14 15:54:00.067147 Waiting for agent ping
3 [Container] 2026/04/14 15:54:00.268355 Waiting for DOWNLOAD_SOURCE
4 [Container] 2026/04/14 15:54:01.546425 Phase is DOWNLOAD_SOURCE
5 [Container] 2026/04/14 15:54:01.547358 CODEBUILD_SRC_DIR=/codebuild/output/src1593886711/src
6 [Container] 2026/04/14 15:54:01.547809 YAML location is /codebuild/output/src1593886711/src/buildspec.yml
7 [Container] 2026/04/14 15:54:01.547932 No commands found for phase name: build
8 [Container] 2026/04/14 15:54:01.547944 Found possible syntax errors in buildspec:
9 In the section BUILD
10 The following keys cannot be identified:
11 cli
12 [Container] 2026/04/14 15:54:01.549524 Setting HTTP client timeout to higher timeout for S3 source
13 [Container] 2026/04/14 15:54:01.549607 Processing environment variables
14 [Container] 2026/04/14 15:54:01.658907 No runtime version selected in buildspec.
15 [Container] 2026/04/14 15:54:01.685679 Moving to directory /codebuild/output/src1593886711/src
16 [Container] 2026/04/14 15:54:01.685718 Cache is not defined in the buildspec
17 [Container] 2026/04/14 15:54:01.728716 Skip cache due to: no paths specified to be cached
18 [Container] 2026/04/14 15:54:01.729093 Registering with agent
19 [Container] 2026/04/14 15:54:01.766067 Phases found in YAML: 1
20 [Container] 2026/04/14 15:54:01.766095 BUILD: 0 commands
21 [Container] 2026/04/14 15:54:01.766392 Phase complete: DOWNLOAD_SOURCE State: SUCCEEDED
22 [Container] 2026/04/14 15:54:01.766407 Phase context status code: Message:
23 [Container] 2026/04/14 15:54:01.873193 Entering phase INSTALL
24 [Container] 2026/04/14 15:54:01.909392 Phase complete: INSTALL State: SUCCEEDED
25 [Container] 2026/04/14 15:54:01.909411 Phase context status code: Message:
26 [Container] 2026/04/14 15:54:01.950040 Entering phase PRE_BUILD
27 [Container] 2026/04/14 15:54:01.950040 Phase complete: PRE_BUILD State: SUCCEEDED
```

### What is being done in this Lab:

- Created the **S3 bucket**
  - Created the **CodeCommit** repository
  - **Uploaded** the **source code** to CodeCommit
  - Added the **build** specification for log analysis
  - Created and linked the **pipeline**
  - Modified the code and **observed pipeline behavior**
  - **Added** an **intentional error** in the YAML file
  - **Reviewed** the **build failure** in the log section
-